Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	699d5facc40b5ad0e9ee5b17
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/

Guidelines:

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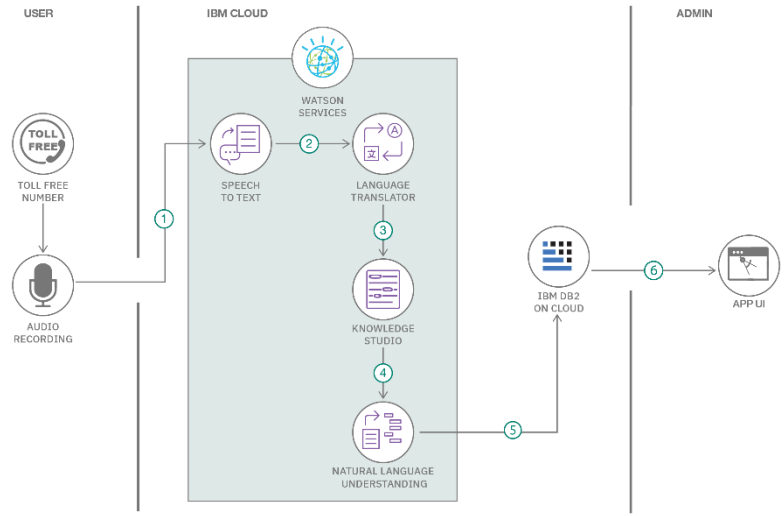

References:

S.No	Component	Description	Technology
1	User Interface	Dashboard interface for viewing electricity consumption trends	Tableau Public
2	Application Logic-1	Script for data cleaning and formatting	Python (Pandas, NumPy)
3	Application Logic-2	Data reshaping and exporting for Tableau	Python
4	Database	Processed CSV/Excel files	Local Filesystem

5	Cloud Database	N/A – Static files used in Tableau	N/A
6	File Storage	Stores preprocessed datasets	Local Storage
7	External API-1	Open energy datasets	data.gov.in, Kaggle
8	External API-2	Weather correlation data (optional)	OpenWeather API
9	Machine Learning Model	Optional forecasting model (future scope)	Prophet / Scikit-learn
10	Infrastructure	Application setup and dashboard hosting	Local preprocessing + Tableau Cloud

S.No	Characteristics	Description / Technology
1	Open-Source Frameworks	Python (Pandas, NumPy), Tableau Public
2	Security Implementations	Tableau's built-in access control and published link privacy
3	Scalable Architecture	Scalable via Tableau's support for large datasets and filters

4	Availability	High availability via Tableau Public's cloud-hosted platform
5	Performance	Fast rendering through preprocessed and optimized data, cache-enabled dashboards

https://c4model.com/ https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/ https://www.ibm.com/cloud/architecture https://aws.amazon.com/architecture https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d